

Zubair Ali Sandhu

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Education

FAST NUCES (National University of Computer and Emerging Sciences)

Aug 2022 – Jun 2026

Bachelor of Science in Computer Science — CGPA: 3.00 / 4.00

Pak Turk Maarif International School and Colleges

Aug 2020 – Aug 2022

Intermediate in Computer Science (ICS)

Relevant Coursework

Generative AI · Computer Vision · Artificial Intelligence · Software Engineering · Mobile App Development · Computer Networks · Database Systems · Data Structures · Object-Oriented Programming

Technical Skills

Languages: Python, C++, Java, Kotlin, C, SQL, JavaScript

Generative AI: Diffusion Models (DDPM), GANs (DCGAN, WGAN-GP, Pix2Pix, CycleGAN), Vision-Language Models (Qwen2-VL), LoRA / QLoRA, Image Translation

ML Frameworks: PyTorch, Hugging Face Transformers, PEFT, OpenCLIP, Scikit-learn, OpenCV, Vision Transformers, Seq2Seq, Contrastive Learning

Web & Backend: FastAPI, REST APIs, Gradio, Streamlit, React, PostgreSQL, Firebase **Mobile:** Android SDK, Jetpack Compose, MVVM, Room

Tools: Git, Hugging Face Hub, Docker, Linux, Kaggle, Android Studio

Projects

Vision to Markdown — Qwen2-VL-2B + QLoRA Fine-Tuning for Document OCR

[Hugging Face](#)

Fine-tuned Qwen2-VL-2B-Instruct for document-to-Markdown OCR using **QLoRA** (4-bit NF4 + rank-16 LoRA on attention projections), trained with SFTTrainer (TRL) on a 350-sample Nougat + LaTeX OCR corpus on Kaggle T4. **Lifted ROUGE-L from 0.41 (zero-shot) to 0.68 (fine-tuned)**; deployed as a Gradio Space with ZeroGPU runtime.

Pixel Genesis — DDPM Image Generator from Scratch in PyTorch

[Hugging Face](#)

Built an end-to-end **DDPM** from scratch — forward Gaussian noising, custom U-Net (64→128→256 channels, sinusoidal time embeddings, GroupNorm residual blocks), and reverse sampling. Trained at 128×128 (**T=300**, β : 1e-4→0.02) on Kaggle T4; deployed as a Gradio Space with intermediate denoising steps.

GANs Verse — DCGAN, WGAN-GP, Pix2Pix & CycleGAN

[Hugging Face](#)

Implemented **four GAN architectures from scratch** in PyTorch. Compared DCGAN vs WGAN-GP on Anime Faces; trained Pix2Pix (U-Net + PatchGAN, $\lambda_{L1}=100$) for sketch colorization and CycleGAN (ResNet-6 + PatchGAN, $\lambda_{cyc}=10$) for unpaired sketch↔photo translation. Shipped **three live Gradio Spaces**.

Neural Storyteller — Image Captioning with Seq2Seq Architecture

[Hugging Face](#)

Built a Seq2Seq captioning system with **ResNet-50 encoder + LSTM decoder**, trained on **Flickr30k** (31K images / 159K captions). Greedy and beam-search ($k=5$) decoding; evaluated with BLEU, METEOR, and ROUGE. Deployed as a switchable Gradio Space.

CRIMEX — AI Crime Reporting & Case Management System

[GitHub](#)

Full-stack AI system spanning Android (Kotlin), React, and FastAPI with PostgreSQL, JWT auth, and SHA-256 evidence hashing. Role-based access control, geo-tagged evidence upload, and AI modules for face recognition, crime hotspot visualization, and embedding-based text pattern analysis.

Multimodal Product Matching — OpenCLIP Retrieval System

[GitHub](#)

Multimodal retrieval system using **OpenCLIP (ViT-B/32)** over the **34K-product Shopee dataset**, jointly leveraging image and text embeddings. Fine-tuned with **Batch Hard Triplet Loss**; evaluated with Recall@K across multiple thresholds.

Experience

Teaching Assistant — Object Oriented Programming

Spring 2025

FAST NUCES, Islamabad

- Mentored **40+ students** on OOP ; ran weekly labs; debugged code; graded assignments and quizzes with consistent rubrics.

Achievements & Certifications

Dean's List — Spring 2023 (SGPA: 3.59 / 4.00), FAST NUCES **The Complete Android 14 & Kotlin Development**

Masterclass — Udemy (Denis Panjuta), *In Progress, expected 2026*